PATRIK SCHULZE

Software Architect | Technical Lead | Senior Full-Stack Engineer

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PROFILE

Programming in the industry since 2008. Software Architect, Technical Lead and Senior Full-Stack Engineer combining broad hands-on engineering expertise with technical leadership and business-oriented product development. Progressed from founding engineer to CTO at a technology startup, helping grow it to over 300,000 registered users while owning architecture, engineering, infrastructure, hiring and cross-functional collaboration. Experienced in building end-to-end systems, integrating AI capabilities into production software, and translating business requirements into practical technical solutions. Technology-agnostic generalist focused on simplicity, maintainability and system-wide architecture, with professional experience in Japan.

TECHNICAL EXPERTISE

Languages & Backend: TypeScript, Node.js, JavaScript, C# .NET, PHP, Java, C++, C, Python, Ruby

Frontend, Mobile & UI: React, Angular, Vue, Qt, OpenGL, Electron, Android, CSS3, SCSS, LESS, Webpack, Vite, pnpm, React Native

DevOps & CI: Linux, Jenkins, AWS, AWS EC2, S3, RDS, Lambda, CodePipeline, AWS CodeDeploy, Route 53, Docker, Bash

Databases: Microsoft SQL Server, Oracle PL/SQL, MySQL, MongoDB, PostgreSQL, Redis

Testing & Automation: JUnit, Puppeteer, Selenium, Cypress, Playwright, Jest

Backend & Web Services: REST, SOAP, Spring, Apache, Express.js

AI ENGINEERING & INTEGRATION

Hands-on experience developing AI-powered applications and assistant systems, integrating LLMs into larger software architectures with application logic, memory, tools and external services.

Models & Platforms: Commercial AI platforms and models from OpenAI, Anthropic and Google; open-weight models including GLM, Qwen, Gemma, Mistral and DeepSeek. Hugging Face, AWS SageMaker, Groq, Fireworks AI, RunPod and other inference providers.

Technologies: LLM integration, agent orchestration, prompt engineering, RAG, embeddings, vector databases, memory management, PyTorch, Whisper, VLMs, voice synthesis and image generation.

Early LLM work: Built a private Amazon review analysis system using Davinci-era GPT-3 models before ChatGPT.

EXPERIENCE

NETCONOMY - Software Architect

NETCONOMY

Vienna, Austria | 2025 – Present

Project: E-commerce storefronts and AI assistants

B2B e-commerce platforms seeking AI-guided customer experiences, AI UX and LLM integrations.

Team Size: 5-10 | **Work Style:** Majority Remote

- Lead end-to-end architecture and delivery for B2B AI-driven e-commerce platforms, turning client requirements into scalable MVPs and proofs of concept.
- Consult directly with stakeholders on project scope, UX/UI specifications and feature sets, aligning business goals with technical execution.
- Coordinate cross-functional frontend, backend and operations teams to preserve architectural integrity across integrations.
- Design AI-guided customer experiences, integrating LLMs and modern web technologies into legacy e-commerce platforms.
- Standardize internal tech stacks and shared libraries across projects, improving delivery efficiency and mentoring junior engineers in Agile Scrum.

OneAct Inc, PieceX - Senior Full-Stack Engineer & CTO

Tokyo, Japan | April 2018 – 2025

Project: E-commerce platform for buying and selling source code

Founding engineer at a startup that grew to over 300,000 registered users. Owned the platform end to end: frontend, design, backend, tooling, DevOps, AWS, hiring and architecture.

Team Size: 5 engineering / 10 company | **Work Style:** 90% Remote

- Progressed from Senior Full-Stack Engineer to CTO in 2022; remained 80% hands-on in programming and technical work, 20% management.
- Built an AI assistant that analyzes customer problems and recommends source-code products, including how to integrate and adapt them.
- Implemented AI solutions using embeddings, GPT, Llama, RunPod, SageMaker and inference APIs, with prompt engineering.
- Architected and maintained full-stack systems, DevOps, tooling and data analysis across web, frontend, backend and command-line tools.
- Administered AWS infrastructure, servers, Docker and Linux; handled systems programming and systems architecture.
- Tech stack: AWS (EC2, RDS, Lambda, S3, IAM, CodePipeline), Node.js, JavaScript, React, Angular, Python, C#, PHP, Docker, Linux.
- As CTO: interfaced with marketing, sales, investors and management; led a 5-person engineering team and coordinated across a ~10-person company.
- Defined company processes, engineering coding guidelines and methodologies.

IBM iX - Full-Stack Engineer

Düsseldorf, Germany | August 2017 – January 2018

Project: Ad and e-commerce web projects for B2B clients

Full-service web projects for B2B clients, covering design, code, backend, hosting and maintenance.

Team Size: 5-10

- Architected and delivered B2B web projects end to end, from frontend to backend, including hosting, maintenance and Jenkins CI.
- Implemented UI/UX, ran client meetings and refined the shared internal tech stack across projects.
- Worked in Agile Scrum; wrote documentation, mentored juniors and handled business travel.
- Tech stack: Gulp, Grunt, Jenkins, Ruby, Java, JavaScript, CSS3, SASS, Spring, Selenium.

Cadosys - C# .NET Engineer

Langenfeld, Germany | June 2016 – April 2017

Project: B2B Windows desktop application for document processing

Windows desktop application that extracts data from documents and reformats it into a controlled, standardized format. Fully remote.

Team Size: 4

- Developed a Windows desktop application in C# .NET using Microsoft SQL Server.
- Improved data algorithms and UI logic; implemented requests for enhancement, database programming and bug fixes.
- Delivered new features, documentation and UX enhancements.

Oracle Maxymiser - Java Engineer

Düsseldorf, Germany | June 2015 – September 2015

Project: User journey flow improvements for B2B solutions

UX journey analysis and improvements for B2B clients and their software solutions.

Team Size: 8

- Worked on Java algorithms against internal databases to transform and migrate data to other systems.
- Built Java tooling for data migration and future use.
- Filtered and processed large datasets; reviewed user journeys and analytics.

Docomo Digital - R&D Software Engineer

Düsseldorf, Germany | March 2013 – March 2015

Project: Telecommunications R&D and prototype development

R&D team inside a telecommunications company, building new prototypes from market and audience research, developing MVPs and talking to customers.

Team Size: 5

- Net mobile AG, owned by NTT Docomo and now called Docomo Digital.
- Developed new prototypes and MVPs for the company.
- Gathered customer feedback to understand audience and market needs.
- Worked autonomously with startup-level ownership in a small team.
- Built every layer of a prototype as a generalist: backend, frontend, tooling, frameworks, UI design and UX flow.
- Forked Firefox OS and ported it to multiple devices for different use cases, including UI/UX.
- Architected a Bluetooth LE advertisement coupon system for mobile.
- Tech stack: Bluetooth, Android, iOS, Objective-C, JavaScript, Gecko, HTML5, CSS3.

University Hospital Research - Full-Stack Software Engineer

Essen, Germany | December 2008 – December 2014

Project: Hospital research department: clinic website and student interface

Full-stack development across clinic needs, from research to website rebuilding and the student platform. Hybrid remote.

Team Size: 3

- Rebuilt the clinic website from scratch and owned its architecture.
- Added analytics and integrations for hospital use cases.
- Built research and scientific tools and handled data work.
- Managed and architected databases and CMS for medical, research and student content.
- Tech stack: PHP, JavaScript, HTML, CSS, Java, Photoshop.

EDUCATION

Düsseldorf University of Applied Sciences

Bachelor of Science in Media Informatics

PERSONAL PROJECTS

- Multimodal, agentic AI assistant integrating speech recognition and synthesis, computer vision, persistent memory, tool execution and 3D avatar animation. Runs locally with system-level access, voice activation and natural conversational dynamics.
- 2D game engine written from scratch in Java, plus 3D games, shaders, animation and physics.

SPOKEN LANGUAGES

English native and preferred, German and basic Japanese